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SARDAR PATEL UNIVERSITY

Master of Science (Information Technology) (Semester 2) Examination 2022

PS02EINT52: Data Science

17/11/2022 Thursday

Time: 10:00 am to 1:00 pm

Marks: 70

NOTE: Figure to the right indicate full marks of the questions.

Q-1 Choose the most appropriate option for each question:

[8]

1. _____ is considered as Model planning tool.
A) SQL Analysis Services B) R
C) SAS / ACCESS D) All of given
2. _____ is a model building tool.
A) SQL Analysis Services B) R
C) WEKA D) All of given
3. In _____ phase of life cycle of data science, there is a need of analytical sandbox in which you can perform analytics for the entire duration of the project.
A) Data Discovery B) Data Preparation
C) Data Modelling D) a None of given
4. Example of _____ data is a heterogeneous data source containing a combination of simple text files, images, videos etc.
A) Structured B) Unstructured
C) Semi-structured D) All of given
5. In the definition of big data, velocity refers to -
A) A mix of many different types of data B) The speed of data generation and processing
C) The large size of data D) None of these
6. Which of the following is not a category in the classification of big data analytics?
A) Descriptive B) Diagnostic
C) Predictive D) None of these
7. The most commonly used data structure in NumPy is -
A) Table B) DataTable
C) ndarray D) DataFrame
8. In R, a variable x can be assigned a value of 10 using -
A) x = 10 B) x <- 10
C) x := 10 D) x == 10

Q-2 Answer the following questions (Any Seven):

[14]

1. Define : Data Science
2. Explain the term Big Data, with example(s)?
3. Explain briefly the characteristics of data Scientist.
4. Write the steps to analyze unstructured data.

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5. Why do we need big data analytics?
6. What is the importance of big data analytics?
7. Explain the arange() method of the NumPy array with an example.
8. Explain the shape() method of the NumPy array with an example.
9. How do we create a list in R? Give an example.

Q-3 Do as directed:

- A. Explain with diagram, main phases of life cycle of Data science. [6]
- B. Discuss shortly the Top THREE data science trends of this decade. [6]

OR

- B. Discuss any SIX Importance of Data Science in Modern Business [6]

Q-4 Do as directed:

- A. Write short note on types of data. [6]
- B. Write short note on characteristics of Big data. [6]

OR

- B. Write short note on Importance / need of Big data. [6]

Q-5 Do as directed:

- A. Explain the meaning and characteristics of big data analytics. [6]
- B. Write a short note on the classification of big data analytics. [6]

OR

- B. Explain the need for and importance of big data analytics. [6]

Q-6 Do as directed:

- A. Describe any 6 operations in NumPy with examples. [6]
- B. Explain the creation of a line chart using Matplotlib. [6]

OR

- B. Write a short note on R. [6]

X
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